

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)	(i)		7.45×10^{-4} mol in 25cm^3 7.45×10^{-3} mol in original solution		1		1		1
		(ii)		moles of HCl in titre B = 1.492×10^{-3} (1) moles of NaHCO_3 in 25cm^3 = $(1.492 \times 10^{-3}) - (7.45 \times 10^{-4})$ = 7.47×10^{-4} moles of NaHCO_3 in original solution = 7.47×10^{-3} (1)		2		2	1	2
		(iii)		moles of $\text{NaHCO}_3 \approx$ moles of Na_2CO_3 therefore $x = 1$			1	1		
	(f)			M_r of anhydrous $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 = 190$ (1) M_r of water present in formula = $226 - 190 = 36$ $\frac{36}{18} = 2$ therefore $y = 2$ (1)		2		2	1	2
				Question 13 total	2	7	3	12	4	11